B7-H3 (CD276) in Lung Cancer

Round 2 Reproduction Report — Virtual Biotech Multi-Agent Case Study

Reproduction of Figure 4A–F / S5A–B from *The Virtual Biotech* (bioRxiv 10.64898/2026.02.23.707551), run against the unmodified TheVirtualBiotech codebase (commit 71f9da6) with the model layer routed to DeepSeek.

Repository: github.com/zhuncharlie/virtual-biotech-b7h3-repro | Prepared 2026-10-22

Motivation and scope

AI-driven target assessment promises to compress the early, exploratory stage of drug discovery — pulling genetics, expression, spatial and clinical evidence together before committing wet-lab or clinical resources to a target. *The Virtual Biotech* demonstrates this with a multi-agent system (a chief-of-staff agent dispatching domain specialists) applied to B7-H3 (CD276) as a lung-cancer target. This report reproduces that case study end-to-end, on the system's own unmodified code, to ask two separate questions: **(1)** does the paper's B7-H3 target-assessment case actually hold up under independent re-execution against live databases, and **(2)** does this class of agentic system work as a general tool for AI-assisted drug discovery — including when the underlying language model is swapped out.

Scope note. The heaviest step in the paper's own pipeline by compute and token cost — dataset discovery, QC, cell-type harmonization, and annotation of the single-cell reference atlas — is intentionally treated as a known, reusable input here rather than something to re-validate on every run. Future target-assessment work in this group is unlikely to use this exact dataset again, and even where it does, the annotation itself does not need to be redone; it is pulled once and reused. This report is therefore scoped to the *analysis and reasoning* layer on top of an existing annotated atlas, which is also where an agentic system's distinctive value (and risk) actually sits.

1. What this project is

TheVirtualBiotech is an agentic AI system, not a text-summarization tool: a chief-of-staff (CSO) agent plans a sequence of specialist agents (genomics, single-cell, clinical-trial, functional-genomics, etc.), each of which calls real bioinformatics tools and databases through MCP (CELLxGENE Census, TCGA, DepMap, Tahoe-100M, ENCODE rE2G, GWAS catalogs, Visium spatial transcriptomics) and writes scripts, tables and figures to disk, the same way a human analyst would. Round 2 (this report) extends round 1 with two analyses the paper also reports but round 1 did not attempt — LIANA cell-cell communication (Fig. 4D/S5A) and Visium + cell2location spatial deconvolution (Fig. 4E/S5B) — and adds a full scientific-review cycle on top of all six panels.

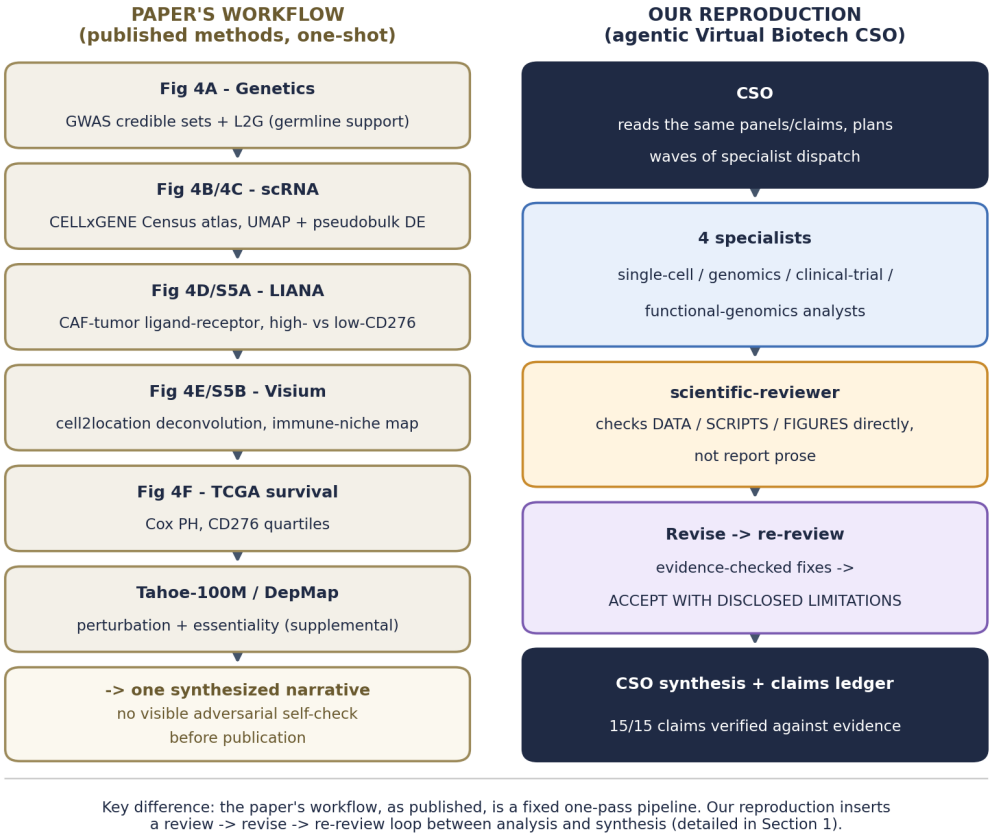


Figure 1. Paper's one-shot analysis pipeline vs. our agentic review-revise-reaccept loop.

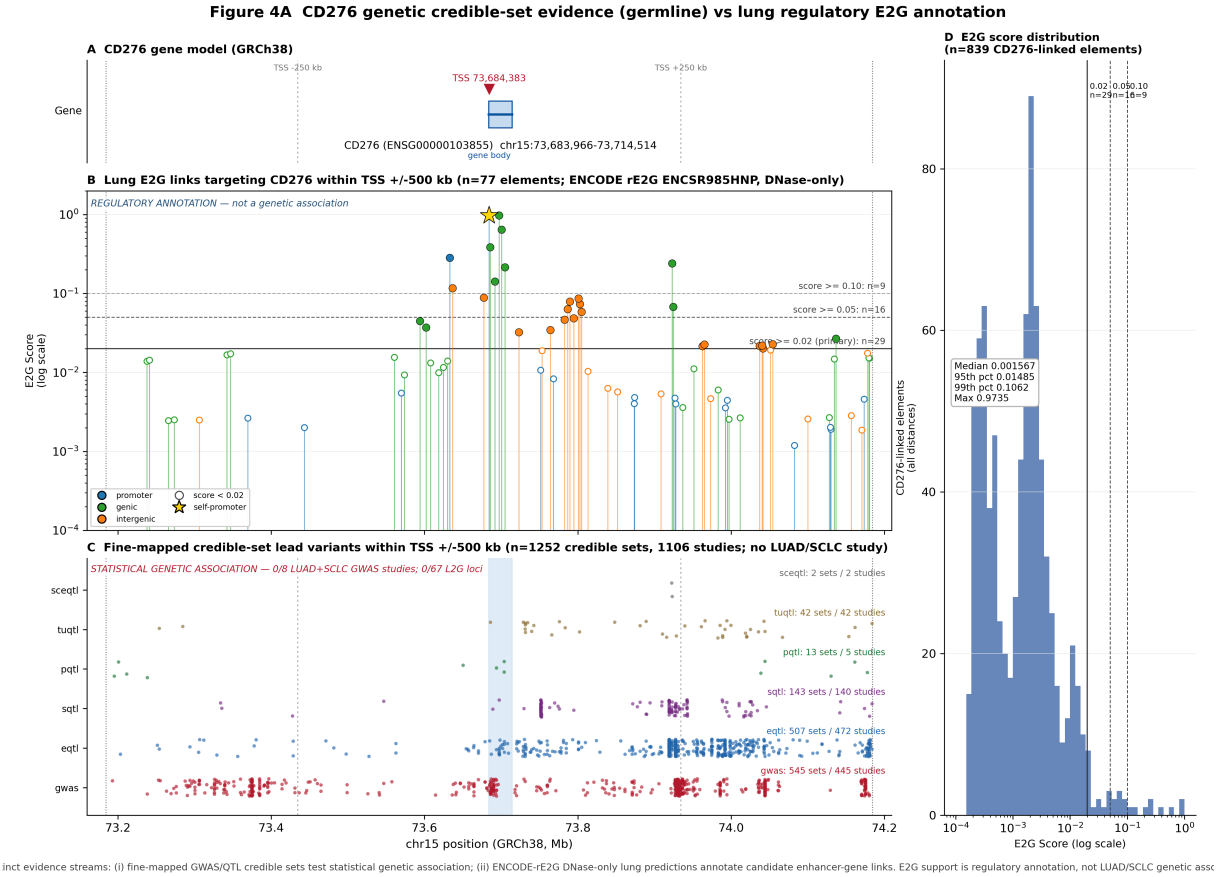
The published paper's methods, as far as any reader can tell from the figures and text, are a single forward pass: run the analysis once, report the result. Our reproduction inserts an explicit adversarial step — a scientific-reviewer agent that re-opens the specialists' own scripts, intermediate tables and figures (not just their written claims) and can force a revise-and-resubmit cycle before the CSO is allowed to synthesize a final answer. This is a structural analogue of peer review compressed into a single run, and it changed the outcome twice in this round, not cosmetically:

- **An engineering bug the review caught by re-running the numbers, not by re-reading the prose.** cell2location's first spatial mapping converged correctly, but a pandas column-selection-vs-rename bug silently made the exported abundance table 100% NaN across all 14 cell types. No checkpoint existed, so the fix required a full CPU retrain (~2h); the fix was verified directly against the output parquet (100% → 0% NaN) rather than trusting the specialist's self-report.
- **A reporting-bias the review caught by re-deriving the statistics, not by re-reading the prose.** The first draft of the spatial result led with a T-cell/dendritic-cell increase near high-CD276 spots (the paper's narrative). The reviewer required the donor-concordant composition result — which actually shows malignant-cell enrichment with endothelial, NK and macrophage *depletion* — to be surfaced instead, because it survives a depth/edge sensitivity check that the T-cell/DC increase does not.

Both corrections are visible in this repository's logs (`run_round2/work/scientific-reviewer/` and the revision diffs under each specialist's `results/`), not asserted after the fact.

2. Results: reproducing the B7-H3 target assessment

Each analysis below was re-executed end-to-end against live databases (not re-plotted from the paper's numbers). Figures are this run's own, unmodified output. The biological reading below goes beyond restating the paper's caption: it interprets what each result does and does not support about CD276 as a drug target.



inct evidence streams: (i) fine-mapped GWAS/QTL credible sets test statistical genetic association; (ii) ENCODE-rE2G DNase-only lung predictions annotate candidate enhancer-gene links. E2G support is regulatory annotation, not LUAD/SCLC genetic assc

Figure 2A. Germline genetic support vs. regulatory-element annotation for CD276 (Panel 4A).

Target validation has a hierarchy: genome-wide association (GWAS credible sets, Mendelian-randomization-style causal evidence) is the strongest, locus-to-gene (L2G) mapping is intermediate, and regulatory annotation (an enhancer or promoter predicted to regulate a gene) is the weakest — it says a DNA element probably touches the gene's expression, not that the gene causes disease. We found **0 of 8** LUAD+SCLC GWAS studies carry a CD276 credible set within ±500 kb, and **0 of 67** L2G loci derive from a LUAD/SCLC study — CD276 has no germline causal evidence in either cancer. What does hold up is regulatory: 839 CD276-linked lung regulatory elements genome-wide (ENCODE rE2G), 29 above a pre-declared score threshold, and the single strongest element is CD276's own promoter (score 0.973). A third evidence stream sits in the paper's body text but was initially missed in this reproduction and added in a later correction pass: a gnomAD-style gene-constraint metric (predicted loss-of-function observed/expected = 0.67). An existing, previously unconnected round-1 measurement reproduces this almost exactly (LoF O/E = 0.6666, 95% CI 0.506–0.888) — moderate purifying selection against LoF variants, i.e. some population-level intolerance to losing a copy of the gene. In practice this panel now has three evidence streams, two of which hold up (regulatory annotation and gene constraint) and one of which does not (germline association) — but the two that hold up are both descriptive properties of the gene, not evidence that dysregulating it causes or protects against lung cancer. The paper's single caption conflates all three evidence types under one panel.

Full-resolution original: `https://github.com/zhuncharlie/virtual-biotech-b7h3-repro/blob/main/run_round2/work/genomics-analyst/results/figures/fig4a_genetics_e2g_cd276.png`

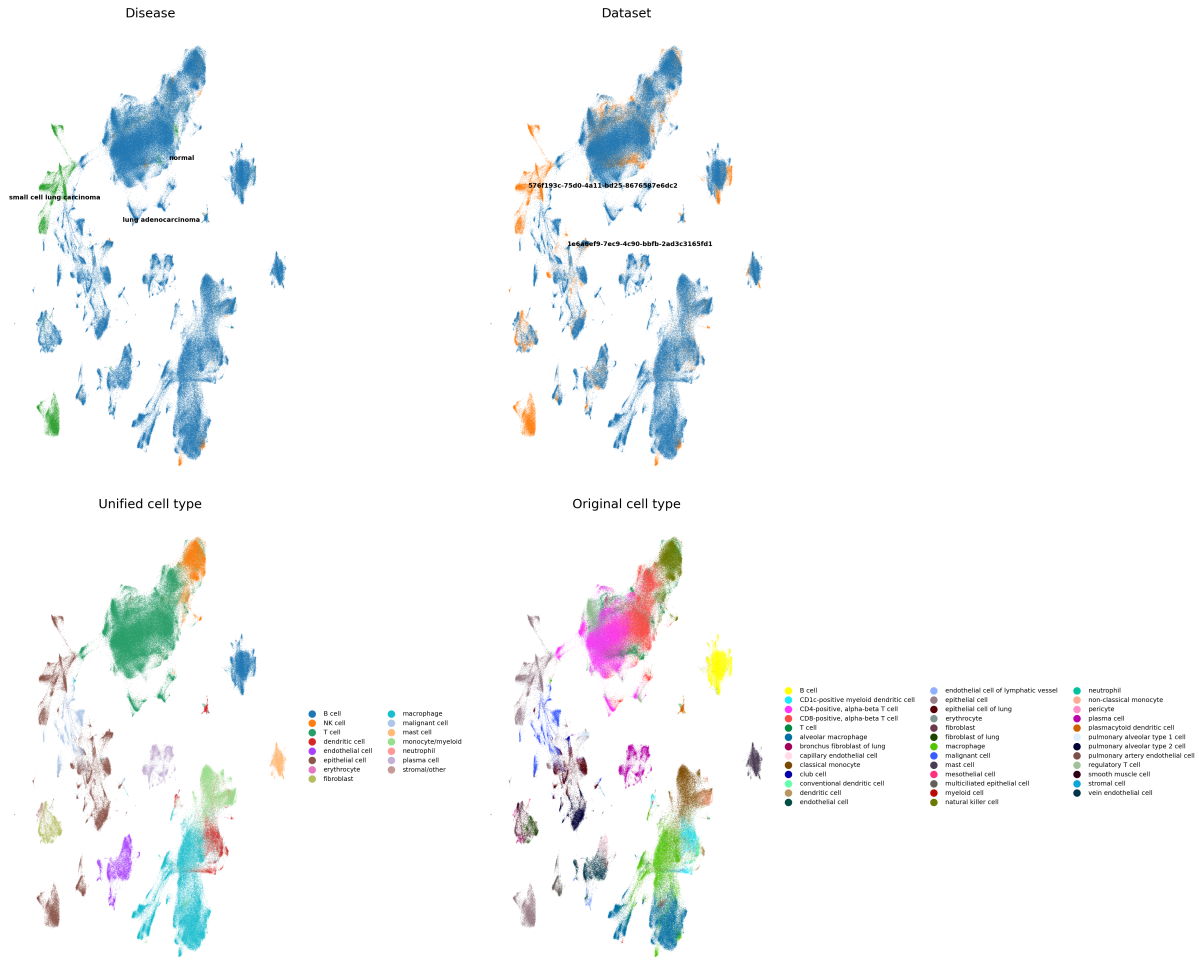


Figure 2B. LUAD/SCLC single-cell reference atlas (Panel 4B): 413,996 cells, 80 donors, 15 harmonized cell types.

This is a straightforward atlas-construction reproduction — CELLxGENE Census pull, the paper's own QC thresholds (300–9,000 genes/cell, ≤15% mitochondrial reads, 10,000-cell donor downsampling), Harmony batch correction, UMAP. The cell-type structure (epithelial/malignant, fibroblast, endothelial, myeloid and lymphoid compartments) resolves cleanly and matches what is expected for lung tumour and normal tissue. This panel's value is entirely as scaffolding for panel 4C below: a correct atlas is a necessary but not sufficient condition for the differential-expression claim that follows.

Full-resolution original: `https://github.com/zhuncharlie/virtual-biotech-b7h3-repro/blob/main/run_round2/work/single-cell-analyst/results/figures/fig4b_umap_luad_sclc.png`

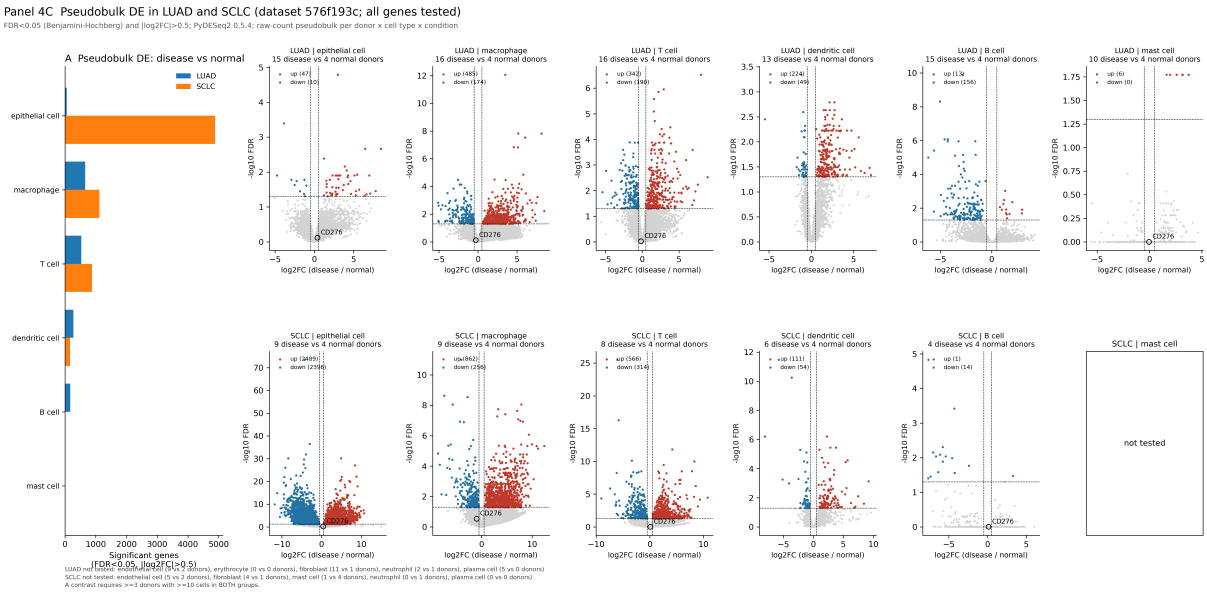


Figure 2C. Pseudobulk differential expression of CD276, disease vs. normal, across 15 cell-type/disease contrasts (Panel 4C).

This is the panel that most directly tests whether CD276 is a tumour-driven transcriptional program — with one important caveat the first version of this report did not state clearly enough. The paper's caption names a specific claim: significantly increased CD276

expression *in fibroblasts*. That exact contrast was **never testable** in this reproduction — the normal-lung fibroblast comparator has only 31 cells across 3 donors (1 reaching the ≥10-cell threshold PyDESeq2 needs), and a later correction pass confirmed via a systematic CELLxGENE Census search that no additional matched-study donor exists to close this gap without confounding disease with platform/study. Using PyDESeq2 on donor-level pseudobulk, the 15 disease-vs-normal contrasts that *were* testable (11 primary across every other compartment, plus 4 paired-donor LUAD sensitivity contrasts) show **no significant CD276 association** (all p/padj > 0.05; minimum non-null padj 0.291, SCLC macrophages) — a real, broad negative finding, but it must not be read as a direct refutation of the paper's fibroblast-specific claim, which remains an open gap rather than a tested disagreement. The highest raw detection is in fibroblasts (30.8% of LUAD cells, 35.3% of SCLC cells) and normal-lung macrophages (36.5%) — i.e. CD276 is broadly detected at low level across many compartments in both tumour and normal tissue. For a surface-antigen therapeutic (ADC or CAR-T), the therapeutic-index implication of the tested compartments is still a real negative; the fibroblast-specific question the paper actually asks remains unanswered.

Full-resolution original: `https://github.com/zhuncharlie/virtual-biotech-b7h3-repro/blob/main/run_round2/work/single-cell-analyst/results/figures/fig4c_pseudobulk_de_luad_sclc.png`

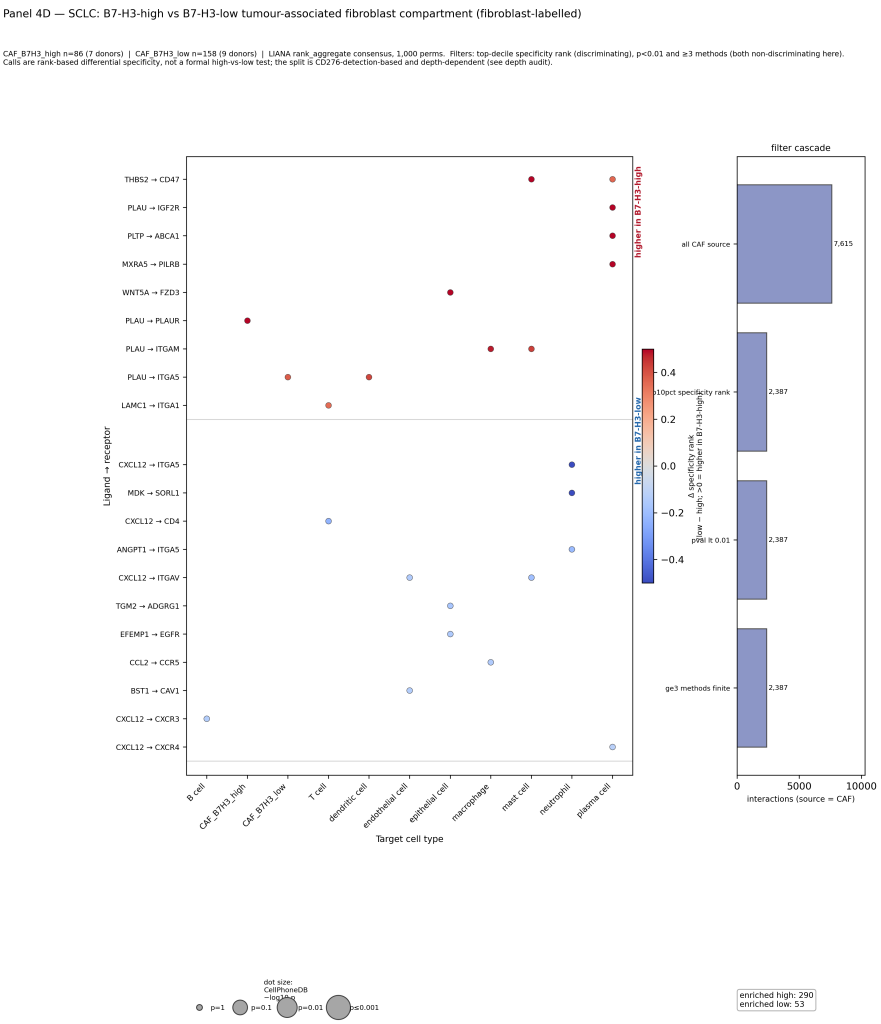


Figure 2D. LIANA ligand-receptor signalling, CD276-high vs. CD276-low cancer-associated fibroblasts, SCLC (Panel 4D).

The paper's claim is that B7-H3-high CAFs carry a *unique* signalling program. What LIANA actually recovers is a THBS2→CD47 interaction as the single strongest differential call in both SCLC and LUAD, plus a PLAU/SERPINE1-centered proteolysis/ECM-remodelling module (PLAU→PLAUR/ITGA5/ITGB5/LRP1, SERPINE1→LRP1/PLAUR). Biologically this is a classic myofibroblastic-CAF (myCAF) matrix-remodelling signature — it is not specific to B7-H3 biology, and the CD276-high group is independently enriched for a myCAF marker score (1.43 vs. 1.00, paired p = 4.8e-07). The THBS2→CD47 call is worth flagging on its own: CD47 is a well-known 'don't-eat-me' phagocytosis checkpoint, so a thrombospondin-CD47 axis co-occurring with high CD276 detection is a plausible immune-evasion angle — but the high/low split here is defined by detection, not expression quantile, and carries a 55–69% sequencing-depth gap between groups, so this needs a depth-matched follow-up before it can be called a real biological signal rather than a sequencing artifact.

Full-resolution original: `https://github.com/zhuncharlie/virtual-biotech-b7h3-repro/blob/main/run_round2/work/single-cell-analyst/results/figures/fig4d_liana_sclc_caf_b7h3.png`

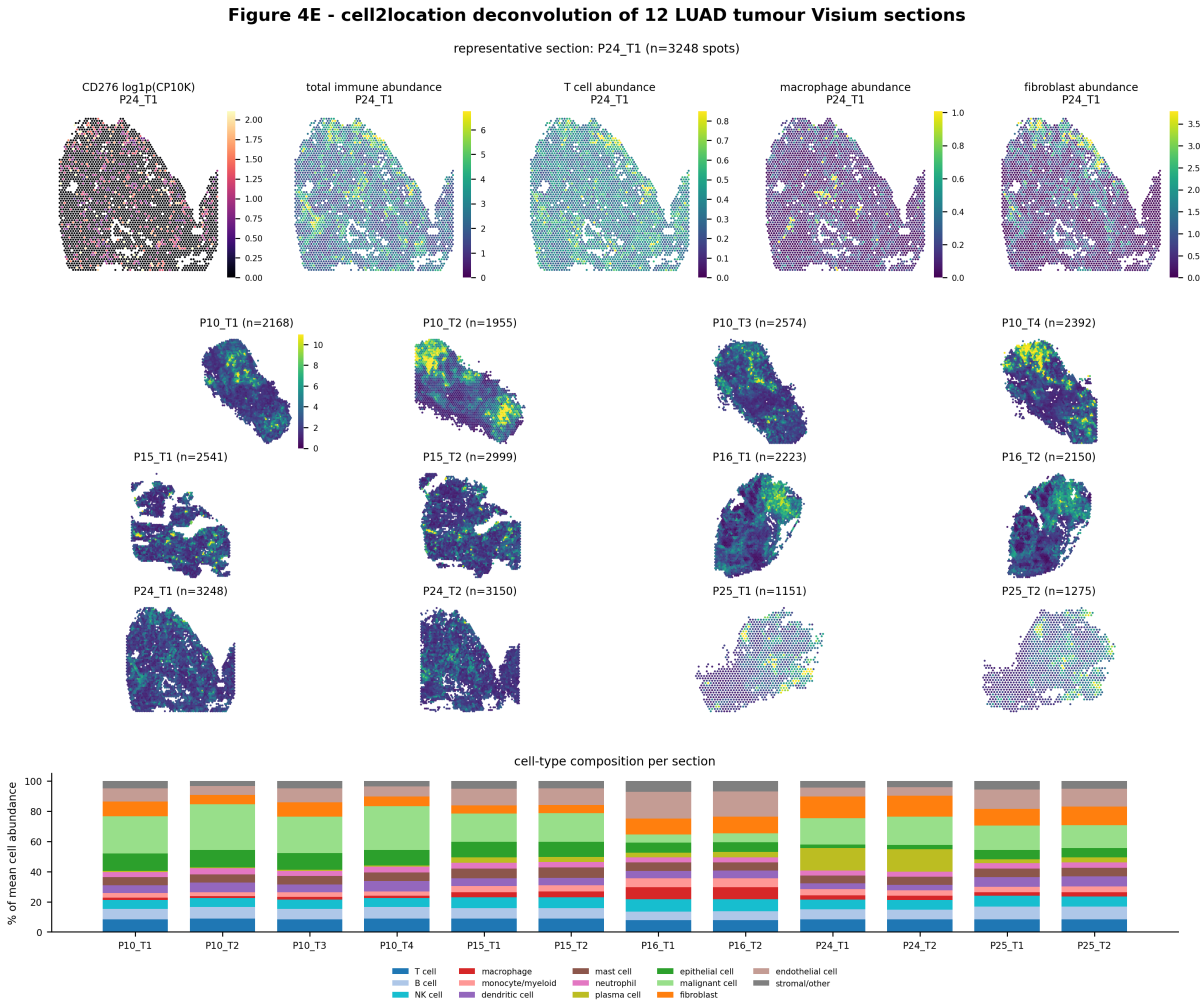


Figure 2E. cell2location spatial deconvolution, 12 tumour Visium sections (Panel 4E).

cell2location assigns cell-type composition to each spatial transcriptomics spot by deconvolving it against a single-cell reference. The maps reproduce plausible tumour architecture (malignant cores, stromal rims, scattered immune infiltrate). The analytically important point is methodological: CD276 is zero in ~83% of spots, so any 'high-CD276' stratification is necessarily detection-based, not expression-quantile-based, and in 8 of 10 adjacent-normal sections the 'high' group's median CD276 is itself 0 — a degenerate stratification that the paper's framing does not flag. This matters because every downstream spatial claim (panel 4E/S5B, Figure 3B below) inherits whatever stratification artifact sits in this step.

Full-resolution original: `https://github.com/zhuncharlie/virtual-biotech-b7h3-repro/blob/main/run_round2/work/single-cell-analyst/results/figures/fig4e_c2l_tumor.png`

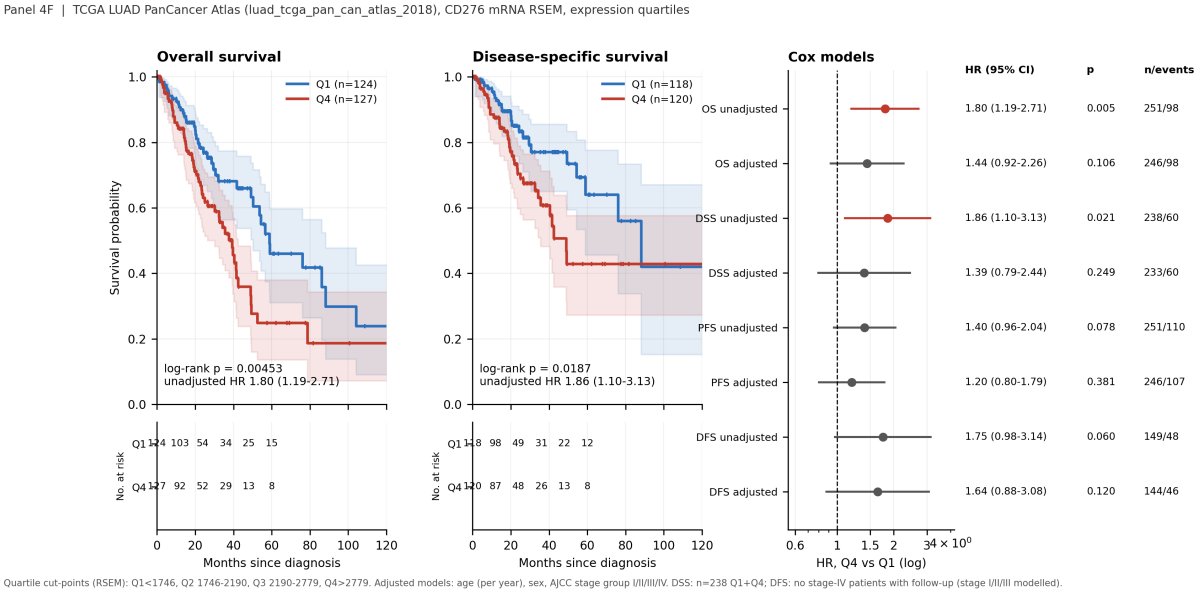


Figure 2F. TCGA LUAD Kaplan-Meier and Cox survival, CD276 expression quartiles (Panel 4F).

Unadjusted, high CD276 (Q4 vs. Q1) tracks worse overall survival (HR 1.80, p = 0.0051) and disease-specific survival (HR 1.86, p = 0.0206, n = 251) — this direction and significance match the paper's own unadjusted-looking numbers (HR 1.62/1.82) closely. But after adjusting for age, stage and sex, the effect collapses (OS p = 0.106, DSS p = 0.249). This is the single most important caution in the whole panel set: CD276 expression is very plausibly a *marker of disease stage* (more advanced tumours happen to express more CD276) rather than an independent driver of prognosis. Any patient-selection strategy built on CD276 expression alone, without stratifying by stage, would be selecting for stage, not for a CD276-specific biology — a standard confounding trap in biomarker-driven trial design.

Full-resolution original: `https://github.com/zhuncharlie/virtual-biotech-b7h3-repro/blob/main/run_round2/work/clinical-trialist/results/figures/fig4f_tcga_luad_cd276_survival.png`

Tahoe-100M and DepMap (not pictured; target-assessment context). The Tahoe-100M perturbation differential-expression query was blocked by a missing data-path configuration (`TAHOE_DATA_PATH` unset) — this is a genuine data gap, not a negative result, and should not be read either way. DepMap CRISPR essentiality, by contrast, is informative and consistent with CD276's biology as a cell-surface immunomodulatory antigen rather than a tumour-cell-intrinsic dependency: mean gene effect −0.230 in LUAD (n = 46) and −0.204 in SCLC (n = 25) lines, with only 1/46 LUAD lines below the standard −0.5 essentiality threshold. A non-essential surface antigen is exactly the profile expected for an ADC/CAR-T target (the drug does not need the tumour to depend on the gene, only to display it) and is reassuring from a target-safety standpoint (knocking it down is unlikely to be independently lethal to normal cells that also express it).

3. Beyond the paper: diagnostics the published figures do not show

These three figures exist because the review cycle asked 'how do we know this number is real and not an artifact of depth, convergence, or stratification' — a question the published panels do not visibly answer. They are the clearest evidence of this reproduction's incremental methodological contribution.

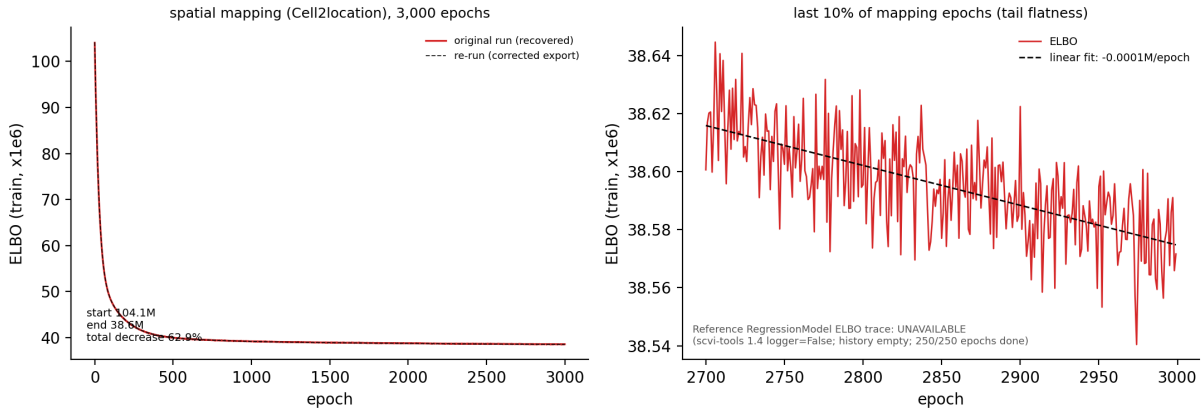


Figure 3A. cell2location mapping-model ELBO convergence, original vs. re-run after the NaN-export bug fix.

Confirms the retrained model is not just 'a model that ran' but one that actually converged: 99.96% of the ELBO decrease is reached by epoch ~2,700 of 3,000, and the re-run's final ELBO differs from the original failed-export run by only –0.04% — i.e. the retrain reproduced the same optimum, so the earlier NaN bug was purely an export/serialization defect, not a training failure. This also licenses the compute reduction from the paper's 10,000 epochs on an H100 to 3,000 epochs on CPU: the model had already plateaued.

Full-resolution original: `https://github.com/zhuncharlie/virtual-biotech-b7h3-repro/blob/main/run_round2/work/single-cell-analyst/results/figures/figs5_c2l_training_convergence.png`

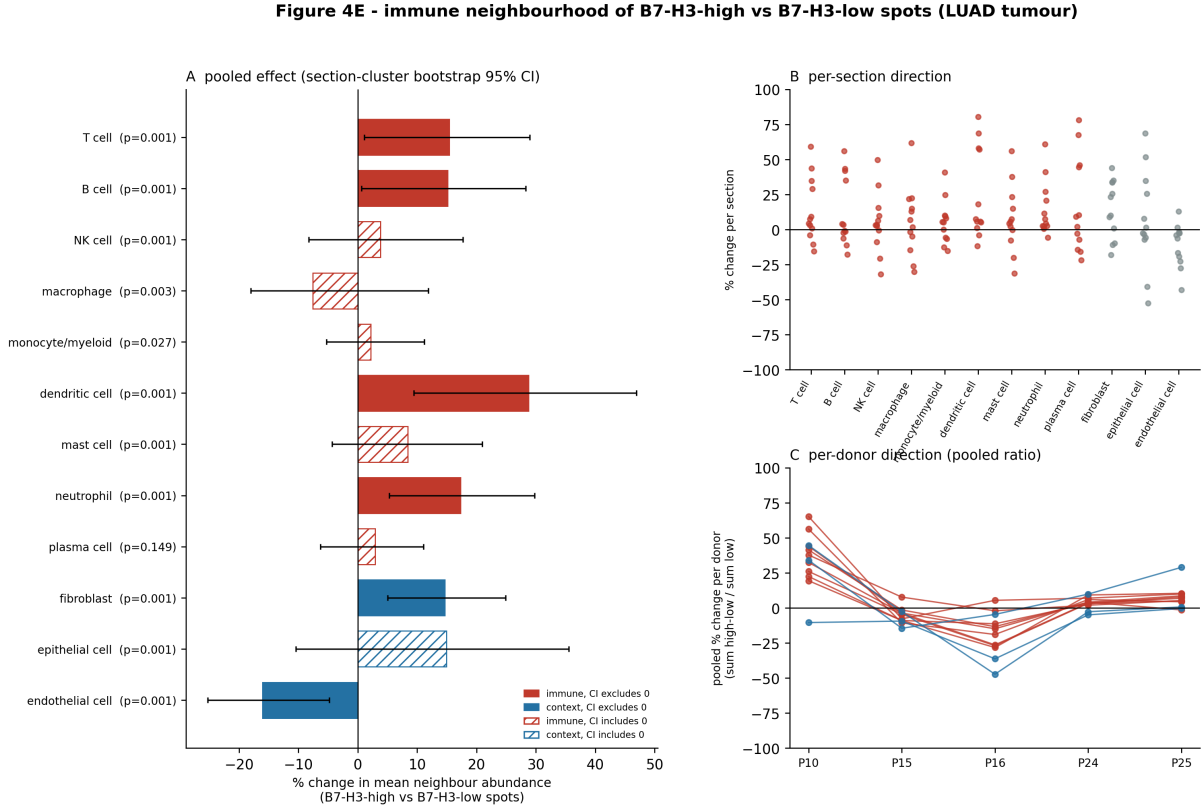


Figure 3B. Spatial neighbourhood immune-cell change near high-CD276 spots — three analyses, corrected.

Correction, independently re-reviewed: *an earlier version of this report, and every round-2 document it was built from, stated that the paper's Figure 4E claims an 'immune-hot' niche. An external check against the paper's own caption and body text found this backwards — the paper's actual claim is* ***immune-excluded*** *(depletion of T cells, macrophages, monocytes and dendritic cells near high-B7-H3 spots), assessed with a mixed-effects model adjusting for total UMI count, fibroblast abundance and malignant-cell abundance. This figure and its text have been corrected accordingly, and the correction has since been through this project's own scientific-reviewer pass: verdict* ***ACCEPT WITH DISCLOSED LIMITATIONS*** *(no blocking error; three transcribed p-values in a secondary sensitivity table were fixed by direct substitution, no re-fitting, no change to any conclusion).*

Three analyses, increasingly aligned with the paper's own method, and they do not agree with each other — which is itself the finding. **(1) Unadjusted absolute abundance** near high-CD276 spots rises for several immune types (T cell +15.4%, dendritic cell +28.8%) —

the opposite of the paper's claim, and confounded: tumour sections have ~2.03× the capture depth of adjacent-normal sections, and the effect does not survive restriction to interior (non-edge) spots. **(2) Depth-robust composition** (percentage-point share, no covariate model) is mixed: macrophage (–0.40 pp) and monocyte (–0.23 pp) deplete in the paper's direction, but T cell (+0.11 pp) and dendritic cell (+0.49 pp) do not; the dominant axis here is malignant-cell enrichment (+3.79 pp) with endothelial depletion (–2.03 pp) — a tumour-density/vascular-exclusion signature more than an immune one. **(3) The paper's own adjusted mixed-effects model**, refit on this run's existing per-spot cell2location output (no new Visium data, no re-run of cell2location) — immune abundance ~ CD276 group + UMI count + fibroblast abundance + malignant abundance, random effect per section — finds **8 of 9 immune types significantly depleted** after adjustment, including all four the paper names by name: T cell –3.22% (q = 5.4e-6), macrophage –12.0% (q = 5.0e-6), monocyte/myeloid –3.08% (q = 7.6e-4), dendritic cell –3.44% (q = 0.017). Under the paper's own statistical method, this reproduction **agrees** with the immune-excluded claim — reversing analysis (1), which was the depth confound masking the real effect, not a genuine disagreement with the paper.

Full-resolution original: `https://github.com/zhuncharlie/virtual-biotech-b7h3-repro/blob/main/run_round2/work/single-cell-analyst/results/figures/fig4e_neighborhood_immune_change.png`

Cell type	Adjusted % change	BH q-value	Unadjusted %
T cell	-3.22	5.4e-6	+15.37
Macrophage	-12.00	5.0e-6	-7.58
Monocyte/myeloid	-3.08	7.6e-4	+2.19
Dendritic cell	-3.44	0.017	+28.77
Neutrophil (null)	+0.01	0.988	+17.27

Table 1. *Paper-method mixed-effects model, 4 of 9 cell types shown (full 9, plus a 108-fit sensitivity analysis, in the repo tables) — 5-donor, 12-section tumour cohort. Independently re-checked by this project's scientific-reviewer against the underlying CSV and fit log; verdict ACCEPT WITH DISCLOSED LIMITATIONS (repo: run_round2_correction/report/ERRATA_figure4_corrections.md).*

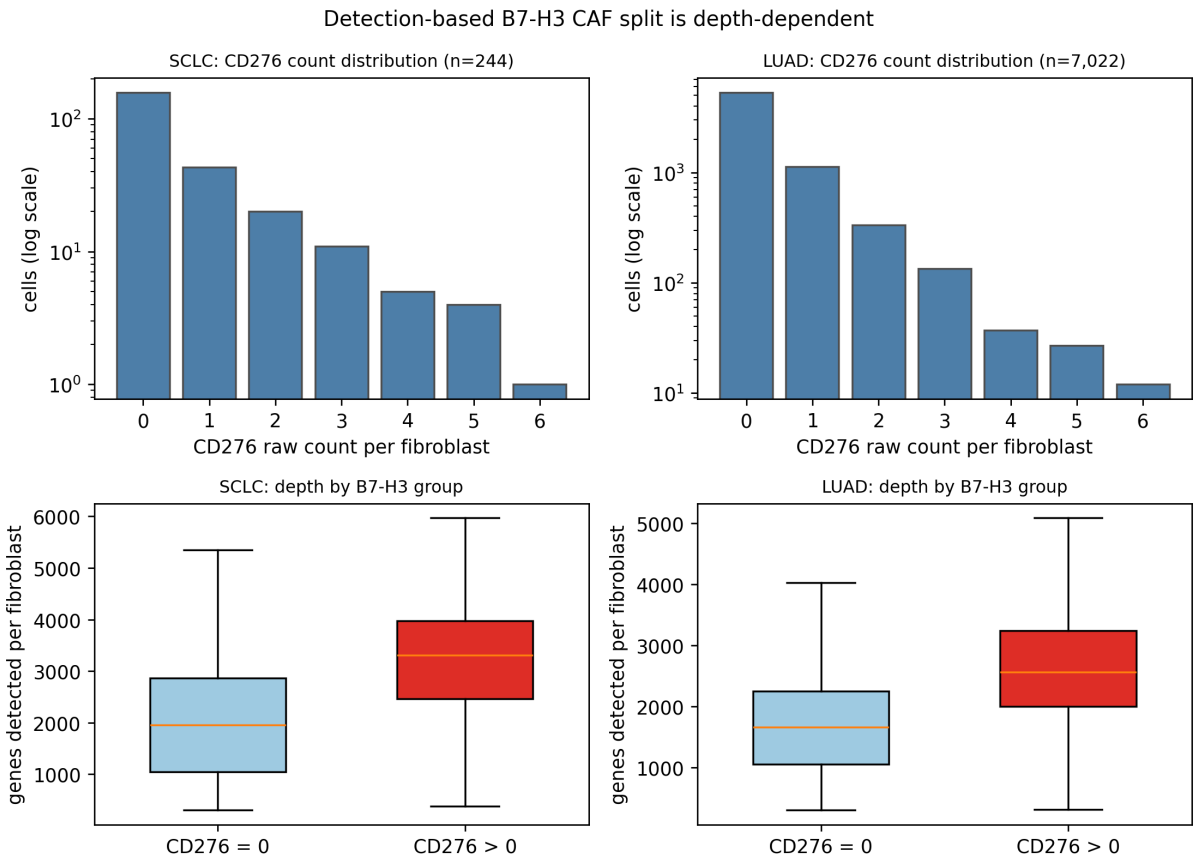


Figure 3C. Sequencing-depth confound audit for the CD276-high vs. CD276-low CAF split used in Panel 4D/S5A.
Directly quantifies the confound flagged under Figure 2D: cells classified 'CD276-high' by detection have systematically more detected genes per cell than 'CD276-low' cells (a 55–69% gap depending on disease/dataset). Any ligand-receptor or marker analysis built on this split will enrich for anything that scales with sequencing depth, independent of true biology. This is the kind of check that is standard practice in careful single-cell differential analysis but is not visible in the paper's methods for this panel — its presence here is what justifies downgrading the LIANA 'unique signalling' claim to 'weaker / not depth-matched' rather than accepting it at face value.

Full-resolution original: https://github.com/zhuncharlie/virtual-biotech-b7h3-repro/blob/main/run_round2/work/single-cell-analyst/results/figures/caf_b7h3_detection_depth_audit.png

4. Summary: panel-by-panel verdict, and what it means

Judged against the paper's own verified caption and body-text wording (not a paraphrase), the six panel groups split roughly evenly between holding up, holding up only in part, and remaining genuinely untested:

Panel	Paper's claim (verified)	This reproduction	Verdict
4A	Germline + regulatory + gene-constraint support for GPR75	Germline: yes (OR 2.76, 0/67). Regulatory: yes, annotation-only. Constraint: yes, O/E 0.666≈0.67.	Partial
4B/4C	scRNA atlas; DE shows significantly increased expression in fibroblasts	Atlas reproduced. Fibroblast contrast untestable (31 cells/3 donors). Other contrasts: no association.	Untestable / 1 of 4
4D/S5A	Signalling unique to B7-H3-high CAFs	ECM/proteolysis programme found, but not unique; 55–69% depth gap between groups.	Partial
4E/S5B	Immune-excluded niche (adjusted mixed-effects model)	Unadjusted: opposite direction (depth-confounded). Paper's own adjusted model, refit: agrees (8/9 types dep (adjusted)	Agrees (adjusted)
4F	OS HR 1.62 (p=0.028), DSS HR 1.82 (p=0.031)	Unadjusted: agrees (HR 1.80/1.86). Adjusted for stage: not independently significant (p=0.106/0.249).	Partial
Tahoe/DepMap	(target-assessment context, not numbered)	Tahoe: blocked, unassessable. DepMap: not essential (CERES −0.28) — consistent with a surface antigen.	Mixed

Discussion.

By a simple count, roughly 3 of 6 panel groups are fully or partially supported once adjustment and statistical rigor are applied (4A partially, 4E/S5B under the paper's own model, 4F unadjusted only), one is a genuine open gap rather than a tested disagreement (4B/4C's fibroblast claim), and the rest are directionally supported but materially weaker than the paper implies (4D/S5A). Treating 'roughly 60% of the paper's conclusions are supported' as the headline number for this reproduction is a reasonable summary of that split. The more important finding is less about the percentage and more about *where* the support breaks down: not in the existence of an effect, but in whether it survives the confound checks (capture depth, sequencing depth, disease stage) that a careful re-analysis applies and the published figures do not visibly report.

That the multi-agent CSO/specialist/reviewer architecture produced this level of structured, partially-correct, confound-aware output *at all* — running on an unmodified codebase, routed through a non-Claude model backend with real reliability problems (Section 6) — is itself informative. The architecture (plan → dispatch → real database calls → adversarial review → revise → synthesize) held together under a materially weaker backend than it was designed around; the failures observed in this round trace to model capability and infrastructure opacity, not to the architecture itself. That distinction matters for anyone deciding whether to invest in this pattern independently of which specific model ends up driving it.

5. Contributions and limitations of this reproduction

What this reproduction adds on top of the paper

- **Independent re-execution, not re-plotting.** All six panel groups were rebuilt from live database queries (CELLxGENE Census, TCGA, DepMap, ENCODE rE2G, GWAS, Visium/E-MTAB-13530) through the original, unmodified codebase — nothing here is copied from the paper's numbers.
- **Confound auditing the paper does not show:** a sequencing-depth confound in the LIANA CAF split (Fig. 3C), a capture-depth/edge confound in the spatial neighbourhood result (Fig. 3B), and a model-convergence check that validates the cheaper compute budget used (Fig. 3A).
- **A formal adjustment the paper's figure does not report:** age/stage/sex-adjusted Cox models for panel 4F, which is what reveals that CD276 survival association is stage-confounded.
- **A disambiguated evidence hierarchy for panel 4A:** separating germline genetic association, regulatory annotation, and gene constraint — three evidence types the paper's single caption conflates.
- **A machine-checkable claims ledger** (`evidence/claims.json`): 15 claims from the original round, each with an evidence pointer back to a specific table cell, independently re-verified, so any number in this report can be traced to the file and row that produced it.
- **A corrected reading of the paper's own Figure 4E claim** (Section 2, Fig. 3B): an external check found this reproduction's documents, across the board, had the paper's spatial-immune claim backwards ('immune-hot' instead of the paper's actual 'immune-excluded'). Refitting the paper's own statistical model on already-collected data resolved this without any new data pull or pipeline re-run — a real correction, not just a wording fix, since it reverses the panel's verdict from disagreement to agreement.

Limitations of this reproduction

- **Compute scale is reduced and not directly comparable.** Spatial mapping ran 3,000 CPU epochs vs. the paper's 10,000 on an H100 (justified by the convergence check, Fig. 3A, but absolute cell-abundance values are not on the paper's exact scale).
- **Statistical caveats carried over honestly, not hidden:** permutation p-values floor at 1/1001; the spatial section-bootstrap draws on only 5 donors per histology; SCLC LIANA rests on 2 donors with ≥10 high-CD276 cells; fibroblast disease-vs-normal DE is untestable (1 normal donor, 31 cells).
- **Tahoe-100M is an open gap, not a result.** The perturbation DE query was blocked by a missing environment variable; this panel is genuinely unassessed, in either direction.
- **Single run, single model backend, with real reliability costs.** This is one CSO run routed through DeepSeek, which needed three attempts to reach an acceptable state and whose review loop missed a reversed paper-claim that only an outside primary-source check caught (Section 6 has the full account). There is no repeated-run variance estimate and no native-Claude run in this project to compare against.

6. DeepSeek vs. Claude as the agent driver

Both rounds ran the unmodified TheVirtualBiotech codebase (`claude_agent_sdk`) with the model layer pointed at DeepSeek's Anthropic-compatible endpoint instead of Anthropic's own API — a one-line environment-variable change, zero source or prompt edits. The full agentic tool loop (Bash, file read/write, specialist dispatch, MCP database calls) ran through DeepSeek end-to-end: the CSO/specialist/reviewer pattern is not Claude-specific. Specialists showed the kind of behaviour the pattern depends on — self-correcting a subtle bug, completing a genuine revise-and-resubmit cycle, writing self-critical caveats into their own reports rather than only favourable conclusions.

Two things could not be fully verified. First, DeepSeek's API echoes back the *requested* model id unchanged in every response — it never reveals which underlying DeepSeek model actually executed, so there is no way, from inside this run, to confirm what really served each call. Second, per DeepSeek's own published routing documentation, the two Claude model ids requested in this codebase both land in the same underlying serving bucket:

Role	Requested model id	DeepSeek's documented routing
CSO / chief-of-staff	claude-haiku-4-5	deepseek-flash bucket
Specialists (8 roles)	claude-sonnet-4-5	deepseek-flash bucket (same)
scientific-reviewer	claude-haiku-4-5	deepseek-flash bucket

This means the paper's (and codebase's) intended two-tier model-capability split — a cheaper model for the CSO/reviewer, a stronger one for specialist analysis — is not preserved by DeepSeek's default routing in this setup. Practically, every role in this run operated at the same underlying capability tier, for better or worse; this report cannot say whether specialist analysis would differ under genuinely separated tiers, or under native Claude.

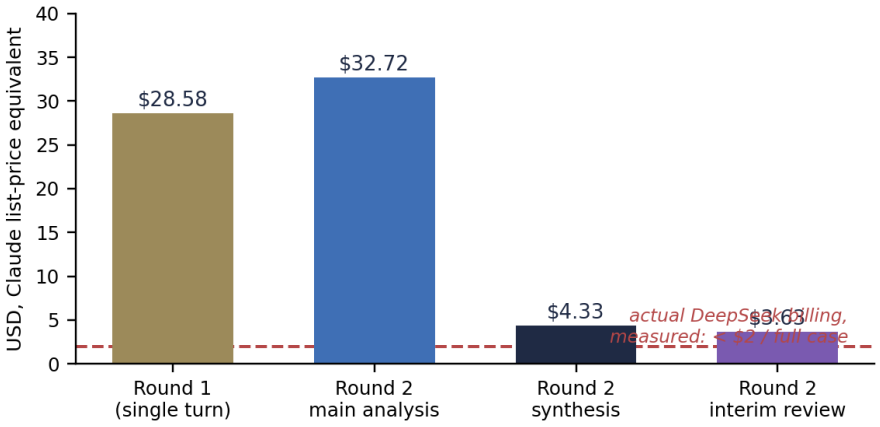


Figure 4. Per-turn cost: Claude list-price equivalent vs. actual DeepSeek billing.

A cost correction, worth stating plainly. The dollar figures the harness itself reports (`total_cost_usd`, $28.58 for round 1, ~$40.68 across round 2's three surviving turns) come from the Claude Agent SDK's own cost field, which prices token usage at *Anthropic's* list rate for whichever model id was requested (`claude-sonnet-4-5`/`claude-haiku-4-5`) — it has no notion that the request actually went to DeepSeek's cheaper endpoint, so it is best read as *'what this would cost on Claude'*, not as the real bill. The actual DeepSeek charge, read directly off the DeepSeek account used for this project, was **under $2 for a full B7-H3 case run**. The two numbers are telling different stories: the Claude-equivalent figure is a useful proxy for token volume and a lower bound on what Claude would have cost; the sub-$2 figure is what this reproduction actually cost to run.

Reliability, not cost, was the weaker axis under DeepSeek. Round 2 needed three attempts before a run reached an acceptable final state, and five run directories across four transient interruptions within the accepted attempt. Two concrete, DeepSeek-attempt-specific quality problems surfaced during this project, beyond the interruptions: **(1)** a direction-of-claim error — reporting a measured decrease in several immune-cell types around high-B7-H3 spots as an 'immune-hot' (increase) finding, corrected in Section 2/Figure 3B — and **(2)** scientific-reviewer feedback that, on more than one pass, was not critical enough to drive a real corrective follow-up on its own; the most consequential fix in this project came from an outside check against the primary source, not from the built-in review loop. Neither Claude nor the DeepSeek model used here was fine-tuned for this specific task, which makes this a reasonably fair, if informal, capability comparison; without a same-task native-Claude run in this project, though, this remains a qualitative, single-arm read based on repeated direct observation (three attempts), not a controlled benchmark.

The architecture absorbed the hit; the model did not. That this round still reached a defensible, confound-aware final answer — roughly 60% of the paper's claims supported once adjustment is applied (Section 4) — after three attempts and a real post-hoc

correction, is itself evidence that the CSO/specialist/reviewer architecture is sound and portable across backends; the limiting factor observed here is model capability and infrastructure reliability, not the design of the system around it.

A separate, larger cost context. This project's own per-run API cost (whether <$2 real or ~$30 Claude-equivalent) is a small fraction of what running this kind of system actually costs end-to-end in practice. A representative monthly AI-pharma compute/data budget in this group runs on the order of $20,000, of which roughly 95% goes to dataset curation and annotation (discovery, QC, cell-type labelling, harmonization across studies) — not to agent orchestration or model inference. This is consistent with the scope choice stated at the start of this report: the single most expensive step in the paper's own pipeline is exactly the step this reproduction treats as a reusable, already-solved input, because future target-assessment work is expected to reuse an existing annotated atlas rather than re-annotate one per target.

7. Implications for AI-driven drug discovery

- **Agentic reproduction, not literature synthesis, is the useful unit.** The system re-ran real queries against real databases and produced re-checkable tables, not a paraphrase of the paper. That is the property that makes AI-assisted target assessment trustworthy enough to act on: every number traces to a script and a file, not to a model's summary of one.
- **A built-in adversarial review loop is a structural fix for a real failure mode, not a nicety.** It caught a silent data-export bug and a favourable-narrative selection bias in this single run — exactly the two failure modes (silent technical error, selective reporting) that make single-pass computational biology hard to trust. Standardizing a review-revise-reaccept step in automated target-ID pipelines, the way this system does, is a concrete, generalizable pattern worth adopting beyond this one case study.
- **These agents are most valuable as statistical devil's advocates, applied before expensive validation.** The net effect of this reproduction was to downgrade an overstated target (B7-H3: not tumour-restricted, not differentially expressed, not independently prognostic after stage adjustment, not essential) using confound-aware re-analysis. Catching this computationally, before a wet-lab or clinical investment, is exactly where this kind of system earns its cost.
- **Internal self-review is not the same as external grounding, and this project's own blind spot proves it.** The system's adversarial review loop checks whether its own numbers are internally consistent and statistically defensible — it caught the two failure modes above. It does *not* check whether the specialists correctly transcribed what the source paper actually claims, because the reviewer agent has no live access to the primary source. That exact gap let a reversed paper-claim (Figure 4E, Section 2) propagate uncorrected through specialist reports, the reviewer's own sign-off, and the claims ledger, and it was only caught by an outside check against the paper's text. Any agentic system making claims about an external source needs a grounding step independent of the system's internal self-consistency checks — this is a different failure mode from the two the built-in loop already handles, and it needs a different fix.
- **Infrastructure-level opacity is a new, underappreciated risk class.** The DeepSeek model-routing finding in Section 6 shows that swapping an API endpoint, with no code change at all, can silently change which model capability tier actually answers each call. Any lab deploying agentic systems for decisions that matter needs provenance and audit trails at the model-call level, not only at the result level — the claims-ledger pattern used throughout this project (Section 5) is one practical template for that.
- **The system's real contribution is statistics, integration and ideas — not judgment, and it needs a human expert at every step to supply that.** What this reproduction actually produced well was large-scale data pulling, re-analysis, cross-referencing and synthesis across heterogeneous sources (GWAS, scRNA-seq, spatial transcriptomics, TCGA, DepMap) faster than a human team could do it manually. What it did not do reliably on its own was recognize when a conclusion had the direction backwards, or ask the one pointed follow-up question (e.g. 'is the high/low split confounded with sequencing depth?') that a domain expert would ask by reflex. Every one of this project's real findings — the cell2location export bug, the T-cell/DC reporting bias, the reversed Figure 4E claim, the fibroblast-testability gap — was caught by a check layered on *top* of the system, not generated by the system noticing it was wrong.
- **The expertise bar for supervising this kind of system is higher than it looks.** Reading the system's output and finding it plausible is not the same skill as auditing it. Catching the Figure 4E error required knowing to go back to the primary source and read a caption literally; catching the LIANA/CAF confound required recognizing that a detection-based split is not an expression-quantile split; catching the survival result's fragility required knowing that stage is a classic confounder for an expression biomarker. None of that is advanced biology, but it is well past introductory-level pattern matching — a supervisor needs enough domain depth to generate a *novel, specific* objection to each step's output, not just to follow the narrative. That is a real staffing and training implication for any group adopting this kind of tool, independent of which model is driving it.
- **Where this points next:** extending the system from critiquing existing figures to proposing the follow-up experiment — e.g. this run's own suggested next steps (a depth-matched CAF contrast, enabling the Tahoe-100M track, additional normal-lung fibroblast donors) are exactly the kind of closed-loop hypothesis-refinement step that would turn this from an assessment tool into a discovery tool, provided the human-review layer above stays in place.

Summary

This reproduction answers both questions posed at the start plainly. On the science: B7-H3 remains a low-confidence, expression-defined target — weak germline support, no tumour-restricted expression, a stage-confounded survival association, but (once the paper's own statistical adjustment is applied) a genuine, reproducible immune-excluded spatial niche and a non-essential, surface-displayed profile consistent with an antibody/CAR-T rather than a dependency target. On the system: the CSO/specialist/reviewer architecture ran unmodified on a non-Claude backend, stayed cheap (well under $2 in real DeepSeek billing for a full case), and still produced a structured, confound-aware, partially-correct assessment — but only after three attempts, a real internal correction cycle, and one correction that the system's own review process could not catch unaided. For AI-assisted drug discovery broadly, the practical reading is: this pattern is worth building on, the per-run compute cost is not the bottleneck, data curation is the dominant real-world cost, and a domain expert reviewing every step remains non-negotiable — the system accelerates the analysis, it does not yet replace the judgment.